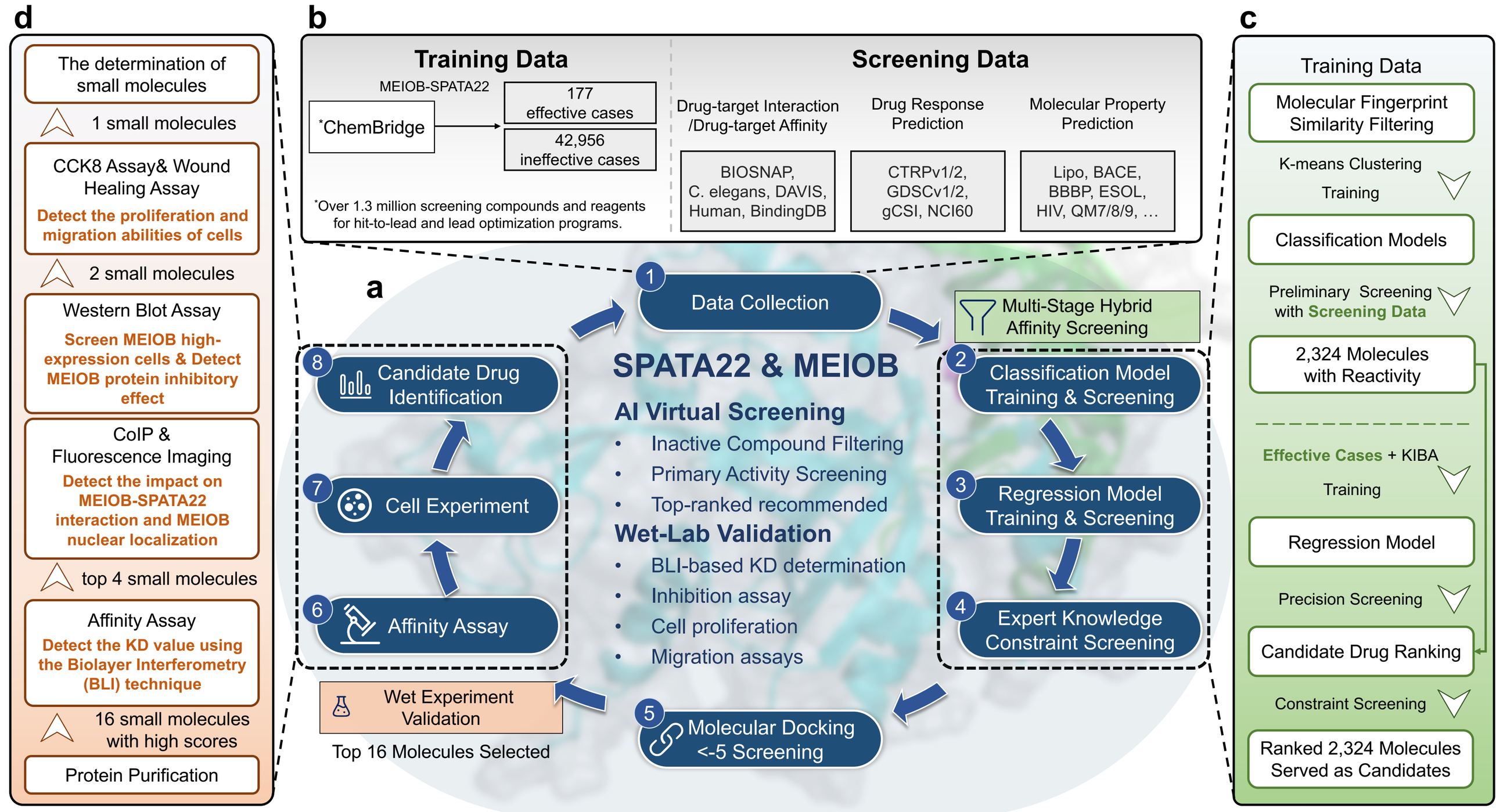

d
The determination of small molecules
1 small molecules
CCK8 Assay& Wound Healing Assay
Detect the proliferation and migration abilities of cells
2 small molecules
Western Blot Assay
Screen MEIOB high-expression cells & Detect MEIOB protein inhibitory effect
CoIP & Fluorescence Imaging
Detect the impact on MEIOB-SPATA22 interaction and MEIOB nuclear localization
top 4 small molecules
Affinity Assay
Detect the KD value using the Biolayer Interferometry (BLI) technique
16 small molecules with high scores
Protein Purification
b
Training Data
MEIOB-SPATA22
*ChemBridge
177 effective cases
42,956 ineffective cases
*Over 1.3 million screening compounds and reagents for hit-to-lead and lead optimization programs.
Screening Data
Drug-target Interaction /Drug-target Affinity
Drug Response Prediction
Molecular Property Prediction
BIOSNAP, C. elegans, DAVIS, Human, BindingDB
CTRPv1/2, GDSCv1/2, gCSI, NCI60
Lipo, BACE, BBBP, ESOL, HIV, QM7/8/9, …
a
1 Data Collection
2 Classification Model Training & Screening
3 Regression Model Training & Screening
4 Expert Knowledge Constraint Screening
5 Molecular Docking <-5 Screening
6 Affinity Assay
7 Cell Experiment
8 Candidate Drug Identification
Wet Experiment Validation
Top 16 Molecules Selected
Multi-Stage Hybrid Affinity Screening
SPATA22 & MEIOB
AI Virtual Screening
Inactive Compound Filtering
Primary Activity Screening
Top-ranked recommended
Wet-Lab Validation
BLI-based KD determination
Inhibition assay
Cell proliferation
Migration assays
c
Training Data
Molecular Fingerprint Similarity Filtering
K-means Clustering
Training
Classification Models
Preliminary Screening with Screening Data
2,324 Molecules with Reactivity
Effective Cases + KIBA
Training
Regression Model
Precision Screening
Candidate Drug Ranking
Constraint Screening
Ranked 2,324 Molecules Served as Candidates